

# PRISCILA DE AZEVEDO DRUMMOND

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## RESEARCH INTERESTS

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How should organizations allocate resources when they act on the same system, their actions interact, and no one is in charge? I build network optimization models for decisions that cross agency and firm boundaries, grounded in structured expert elicitation and administrative data, to show when coordination is worth its cost.

Coordinated decision-making across organizations · Supply Chain Management · Prescriptive analytics for multi-stakeholder intervention · Resource allocation in constrained service and supply systems · Supply chain compliance and labor risk · Measurement design for contested and multi-dimensional outcomes · Responsible analytics and the consequences of optimized decisions · Analytics and operations pedagogy

**Methods:** network optimization and interdiction · integer and mixed-integer programming · Bayesian decision analysis and cost-benefit modeling · multilevel and hierarchical regression · data envelopment analysis · expert elicitation · systematic and scoping review · qualitative and participatory methods

## EDUCATION

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**Northeastern University**, Boston, MA

Ph.D., Industrial Engineering, expected December 2026 (defense scheduled 2 December 2026)

Dissertation: *Collaborating Agents and Network Interdiction Models*

Advisor: Dr. Kayse Lee Maass. Committee: Dr. Shahin Shahrampour, Dr. Gabriela Gongora Svartzman

**Northeastern University**, Boston, MA

Certificate, Post-secondary Teaching, Spring 2026

Capstone: Course design for *System Engineering in Public Programs* (graduate level) and *Deterministic Operations Research* (graduate level)

**University of Texas Rio Grande Valley**, Edinburg, TX

M.S., Engineering Management, 2021

Thesis: *A Framework to Identify Human Trafficking at the U.S. Border* (Advisor: Dr. Hiram Moya)

**Universidade Federal do Rio Grande do Sul**, Porto Alegre, Brazil

M.S., Business Administration, 2019

Thesis: *Relative Efficiency Analysis of the Women's Crisis Center Network in Rio Grande do Sul* (Advisor: Dr. Denise Lindstrom)

**Universidade do Estado do Rio de Janeiro**, Rio de Janeiro, Brazil

M.P.S., Finance and Planning, 2016

Project: *Financial and Sporting Efficiency of Brazilian Soccer Clubs* (Advisor: Dr. Francisco Pessanha)

**Universidade Federal do Rio de Janeiro**, Rio de Janeiro, Brazil

B.A., Industrial Design, Product Design, 2011

Project: *Trebuchet: A Multidisciplinary Pedagogical Game* (Advisor: Dr. Roosevelt Telles)

## REFEREED JOURNAL ARTICLES

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Jafari, A., **Drummond, P. A.**, Bhimani, S., Nishigaya, D., Ding, A. A., Farrell, A., & Maass, K. L. (2024). Enhancing detection of labor violations in the agricultural sector: A multilevel generalized linear regression model of H-2A violation counts. *PLOS ONE*, 19(5), e0302960. <https://doi.org/10.1371/journal.pone.0302960>

## MANUSCRIPTS UNDER REVIEW

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Jafari, A., Bhimani, S., **Drummond, P. A.**, Farrell, A., & Maass, K. L. Network interdiction to improve labor trafficking detection in the U.S. agricultural sector. Revise and resubmit at *Decision Sciences*.

## REFEREED CONFERENCE PROCEEDINGS

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**Drummond, P. A., & Moya, H.** (2020). U.S. ports of entry's decision model with an anti-human trafficking focus. *Decision Sciences Institute Proceedings*. ISSN 2471-884X

## WORKING PAPERS

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**Drummond, P. A., & Maass, K. L.** Multiple attacks in network interdiction problems: A scoping review. Status: revision complete. Target: *Networks* (submitting September 2026).

**Drummond, P. A., & Maass, K. L.** Collaboration in anti-trafficking efforts: A network interdiction problem with multiple dependent attacks. Status: results analysis. Target: *Production and Operations Management* (September 2026). **[Job Market Paper]**

**Drummond, P. A., & Maass, K. L.** Optimal bounds and complexity for network interdiction with multiple interventions. Status: results analysis. Target: *Operations Research*.

**Drummond, P. A., & Maass, K. L.** Algorithms for network interdiction problems with non-convex functional attacks. Status: model and algorithm ready; coding. Target: *European Journal of Operational Research*.

## RESEARCH IN PROGRESS

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**Drummond, P. A., & Maass, K. L.** Unintended consequences of enforcement in labor trafficking networks: geographic displacement, operational fragmentation, and compliance switching. Status: ideation and literature review stage; resuming after degree completion. Target: *Manufacturing and Service Operations Management*

**Drummond, P. A.** Teaching fellowship impact on industrial engineering teaching education - An auto-ethnography account by a student. Status: data collection, cleaning, and initial literature review ready; resuming after degree completion. Target: *Journal of Engineering Education*

## CONFERENCE PRESENTATIONS

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### \*PRESENTER

**Drummond, P. A.\*, & Maass, K. L.** (2026, November). Network interdiction model with multiple dependent attacks, job market talk. INFORMS Annual Meeting, San Francisco, CA.

**Drummond, P. A.\*, & Maass, K. L.** (2025, October). Collaboration in anti-trafficking efforts: A network interdiction problem. INFORMS Annual Meeting, Atlanta, GA.

**Drummond, P. A.\*, Jafari, A., & Maass, K. L.** (2024, October). Modeling strategies to disrupt labor trafficking. INFORMS Annual Meeting, Seattle, WA.

**Drummond, P. A.\*, Jafari, A., & Maass, K. L.** (2023, October). Labor trafficking disruption with collaborating agents. INFORMS Annual Meeting, Phoenix, AZ.

Jafari, A.\*, **Drummond, P. A.**, Bhimani, S., Farrell, A., & Maass, K. L. (2023, October). Breaking the chains: An innovative network interdiction approach to combat labor trafficking in agricultural supply chains. INFORMS Annual Meeting, Phoenix, AZ.

**Drummond, P. A., Jafari, A., Childress, C., & Maass, K. L.** (2022, November). Leveraging public data from illicit activities in operations research. Decision Sciences Institute Annual Conference.

**Drummond, P. A., & Maass, K. L.** (2022, October). International migration and human trafficking networks. INFORMS Annual Meeting

Jafari, A.\*, **Drummond, P. A.**, Nishigaya, D., Bhimani, S., Farrell, A., & Maass, K. L. (2022, October). Labor violation trends in H-2A nonimmigrant agricultural workers: Improving inspection strategies for detecting labor violations. Society for Risk Analysis Annual Meeting.

Gurrola, N.\*, Hallas, K.\*, **Drummond, P. A.\*, Rodriguez, A.\*, & Cano, S.\*** (2021, April). Understanding female STEM identity at UTRGV through narrative inquiry. Engaged Scholar Symposium, University of Texas Rio Grande Valley, Edinburg, TX. Recording: <https://www.youtube.com/watch?v=8gJWmTL9PK0>

**Drummond, P. A.\*, & Moya, H.** (2021, April). Optimization of port-of-entry operation in the U.S.: An anti-human trafficking focus. 5th Coastal Bend Mathematics and Statistics Conference, Texas A&M University, College Station, TX.

- Drummond, P. A.\*, & Moya, H.** (2020, November). U.S. ports of entry's decision model with an anti-human trafficking focus. Decision Sciences Institute Annual Conference, Dallas, TX (virtual).
- Drummond, P. A.\*, & Moya, H.** (2020, November). A framework to identify human trafficking at the U.S. border. IISE Annual Conference and Expo, New Orleans, LA (virtual).
- Drummond, P. A.\*, Rahal, J., & Amorim, F.** (2008, June). Analysis of different educational strategies using work-based pedagogy at the middle school Colegio de Pescadores, Macae, RJ. 5o Congresso de Extensao da UFRJ, Rio de Janeiro, Brazil.

## POSTER PRESENTATIONS

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- Drummond, P. A.\*, & Maass, K. L.** (2024, March). When to collaborate? Labor trafficking interdiction with multiple agents. RISE, Northeastern University, Boston, MA.
- Drummond, P. A.\*, & Maass, K. L.** (2023, November). Labor trafficking disruption with collaborating agencies. MIE Research Expo, Northeastern University, Boston, MA.
- Drummond, P. A.\*, & Moya, H.** (2021, February). A decision support tool at the U.S. port of entry with a human trafficking focus. Engineering Week, University of Texas Rio Grande Valley, Edinburg, TX.
- Drummond, P. A.\*, & Moya, H.** (2020, February). A framework to identify human trafficking at the U.S. border. Engineering Week, University of Texas Rio Grande Valley, Edinburg, TX.
- Drummond, P. A.\*, & Moya, H.** (2019, November). A framework to identify human trafficking at the U.S. border. Graduate Research Fair, University of Texas Rio Grande Valley, Edinburg, TX.

## RESEARCH EXPERIENCE

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### **ORSJ, Northeastern University, Boston, MA**

*Doctoral Researcher, 2021 to present*

Developed network optimization models for multi-agency coordination against labor trafficking, including a new formulation for dependent interventions on a shared network. Cleaned and analyzed three public datasets (500,000+ records) in Python to study labor violation patterns and inspection strategy. Contributed to a multilevel regression study of H-2A violations with an interdisciplinary team spanning criminal justice, social service, engineering, and business. Elicited intervention probabilities from anti-trafficking specialists for use as model parameters. Translated qualitative data to useful forms for Operations Research models (federal cases). Partners: U.S. Department of Homeland Security CINA Center; criminal justice researchers at Northeastern, legal advocacy organization.

### **University of Texas Rio Grande Valley, Edinburg, TX**

*Lead Researcher, Female Students' Perception of STEM Identity, October 2020 to June 2021*

Originated the study and secured IRB approval; led a five-member undergraduate research team. Recruited and interviewed 11 participants using a semi-structured protocol. Analyzed first-phase data using narrative inquiry with a priori and emergent coding and member checking, applying narrative inquiry and critical-consciousness frameworks. Disseminated findings at the Engaged Scholar Symposium and the RGV STEM Education Consortium (2021); co-authors continued the study after 2021, and it was published as Hallas & Chapman (2024), *RGV STEM Ed Conference Proceedings*, <https://doi.org/10.51734/yggi4440>

*Graduate Researcher, 2019 to 2021*

Built a Bayesian decision model integrating cost-benefit analysis and statistical risk predictors to determine screening thresholds for trafficking detection at U.S. ports of entry.

### **Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil**

*Graduate Researcher, 2017 to 2019*

Partnered with the Rio Grande do Sul Public Security State Department to evaluate resource allocation across the network of services for survivors of domestic violence. Mapped multi-service provider flows, survivor pathways, and information pathways through the system. Conducted data envelopment analysis on crime and social service data across 450+ municipalities using R, combined with structured expert validation. Found that the range of services available constrained effectiveness more than total investment.

### **Universidade do Estado do Rio de Janeiro, Rio de Janeiro, Brazil**

*Graduate Researcher, 2015 to 2016*

Applied data envelopment analysis to 42 Brazilian soccer clubs, benchmarking financial inputs against sporting performance to identify inefficiency in spending and investment.

**Universidade Federal do Rio de Janeiro**, Rio de Janeiro, Brazil

*Research Coordinator, STEM Pedagogy Assessment, 2006 to 2009*

Coordinated a faculty group assessing the outcomes and challenges of a learning-by-working pedagogical model at a project-based public middle school serving a gang-controlled community in Macae, RJ. Qualitative design; results presented at a university extension conference in 2008.

## GRANTS AND COMPETITIVE FUNDING

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- **Dissertation Completion Fellowship**, Northeastern University, 2026
- **Teaching Fellowship**, Northeastern University, 2025 (yearly student-awarded fellowship)
- **Dean's Chair Fellowship**, Northeastern University, 2021
- **Presidential Graduate Research Award**, UTRGV, 2019
- **CNPq Research Award**, 2017 (Brazilian competitive research grant and scholarship)
- **Travel Funds**, PhD Network, Northeastern University, 2022, 2023, 2024, 2025

## HONORS AND AWARDS

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- **College of Engineering Outstanding Graduate Teaching Award**, Northeastern University, 2026 (college-level award)
- **Tau Beta Pi**, Engineering Honor Society, member since 2021
- **Outstanding Student Award for Academic Achievement**, UTRGV, 2020

## TEACHING EXPERIENCE

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### Instructor of Record

*Northeastern University, Boston, MA, Spring 2026*

OR 6205: *Deterministic Operations Research* (graduate, 4 credits, 6 enrolled), Spring 2026

Full responsibility for course design, instructional materials, in-class active learning exercises, assessment design, lecturing, grading and feedback, and project mentoring.

Student evaluations rated above departmental means for instructor enthusiasm, feedback sufficiency, an inclusive learning environment, and organization of course materials, and above departmental means for the online learning environment on organization of weekly materials, connection with the instructor, and community among classmates; full institutional course evaluation report in the teaching portfolio: <https://pridrum.github.io/teaching/>

### Graduate Teaching Assistant

*Northeastern University, Boston, MA, 2023 to 2026*

- *IE 7200: Supply Chain Engineering* (graduate), Summer II 2026, Prof. A. Guschin, 11 students
- *IE 6200: Engineering Probability and Statistics* (graduate), Summer I 2026, Prof. Doug Schmucker, revision materials
- *IE 5617: Lean Concepts and Applications* (undergraduate and graduate). Summer II 2026, Prof. A. Guschin, 22 students; Summer I 2025, Prof. H. McManus, 20 students; Summer II 2023, Prof. A. Guschin, 46 students; Summer I 2023, Prof. H. McManus, 45 students
- *IE 4516: Quality Assurance* (undergraduate), Fall 2023, Prof. H. McManus, ~20 students, course design

Taught the Process Control and Six Sigma Basics module and the Quality Tools and Topics module independently (Summer I 2025). In the end-of-term student ranking of all 17 class sessions, these placed 3rd and 7th, at 4.81 and 4.67 on a 5-point scale, above the instructor's own scores for the same sessions.

For IE 4516, led a complete redesign of the course teaching materials with Prof. McManus and Dr. Rehab Ali, for a course built around experimental, simulation-based, and project-based learning. Refactored and delivered two modules: Design of Experiments and ANOVA, including the classroom game I designed for it, and Process Control.

For IE 6200, analyzed the materials and assignments against AI usage, reviewed material from a student perspective, and created answers with different prompts using AI to test robustness of assignments to use of artificial intelligence. Weekly meetings with the professor, and a feedback report for all course materials.

Across all Northeastern appointments: developed supporting instructional materials and interactive exercises, held office hours, supported weekly recitations, and graded homework and projects.

*University of Texas Rio Grande Valley, Edinburg, TX, 2019 to 2021*

- *Engineering Management* (undergraduate), Summer 2021, Prof. H. Moya, 20 students
- *Manufacturing Production and Control* (undergraduate), Fall 2020, Prof. H. Moya, 18 students
- *Operations Research* (undergraduate), Spring 2020 and Spring 2021, Prof. H. Moya, 21 students

Created targeted tutorial sessions and instructional videos during the pandemic transition; implemented personalized feedback systems and supported weekly recitations.

*Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil, 2017 to 2019*

- *Manufacturing Production and Control*, Fall 2017, Spring 2018, Spring 2019, Prof. D. Lindstrom, 30+ students per term

Delivered bi-weekly research seminars and provided project feedback for industry-partnered student work.

### **Prior Teaching and Curriculum Leadership**

*Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil, 2006 to 2009*

Middle School Teacher and Pedagogical Project Coordinator, 90+ students per year

Coordinated the engineering, design, and technology curriculum for a project-based learning school. Managed an interdisciplinary team of 10 to 15 undergraduate interns. Designed and taught experiential learning modules integrating engineering principles and design for students aged 11 to 16 through boat design and construction.

### **COURSES PREPARED TO TEACH**

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**Ready to teach immediately.** Operations Management · Supply Chain Management · Supply Chain Analytics · Operations Research and Deterministic Optimization · Prescriptive Analytics · Introduction to Data Analysis · Lean Operations and Process Improvement · Quality Management · Engineering Economy and Engineering Economics and Finance · Project Management

**Ready with one semester of preparation.** Predictive Analytics · Data Visualization · Simulation Modeling · Business Analytics · Decision and Risk Analysis · Sustainable and Responsible Supply Chains · Python for Business Analytics

**Ready to develop.** Operations Management and Social Impact · Sports and Performance Analytics

### **CURRICULUM DEVELOPMENT**

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- Course design and instructional materials for graduate level course, *OR 6205: Deterministic Operations Research*, Northeastern University, Spring 2026. New materials, in-class active learning exercises, assessments, and negotiated syllabus components.
- Course design, *System Engineering in Public Programs*, post-secondary teaching certification capstone, Northeastern University, 2026.
- Complete redesign of teaching materials (with Prof. McManus and Dr. Rehab Ali), *IE 4516: Quality Assurance*, Northeastern University, 2023. Brought the existing materials together and set the sequence, overarching topics, and objectives for a course built around experimental, simulation-based, and project-based learning. Refactored and delivered two modules: Design of Experiments and ANOVA, including the classroom game I designed for it, and Process Control.
- Video tutorial series for Operations Research during the remote-teaching transition, UTRGV, October 2020 to April 2021 (<https://youtu.be/GOG1Yf72Pow> · <https://youtu.be/gbPIL9ZMawA> · <https://youtu.be/ISAcgCrVY8w> · <https://youtu.be/rQ6V6cQWGy8>).
- Engineering, design, and technology curriculum, project-based middle school, Universidade Federal do Rio de Janeiro, 2006 to 2009.

## TEACHING DEVELOPMENT AND PEDAGOGICAL TRAINING

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Post-secondary Teaching Certification, Northeastern University, 2025 to 2026

Teaching Fellowship (mentored teaching, *Deterministic Operations Research*), Northeastern University, Fall 2025 to Spring 2026

Future Business Analytics Professors Workshop, Tippie College of Business, University of Iowa, 2026

Teaching workshop, INFORMS Committee on Teaching and Learning, 2025

## ADVISING AND MENTORING

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- Mentoring a master's student in an independent research project (Fall 2026, in progress).
- Mentored five graduate student projects (2 doctoral, 3 master's, drawn from several engineering disciplines) in OR 6205: *Deterministic Operations Research*, Northeastern University, Spring 2026. Project topics included trip scheduling and ranking.
- Led and mentored a five-member undergraduate research team as lead researcher, STEM identity narrative inquiry study, University of Texas Rio Grande Valley, 2020 to 2021; team members went on to graduate school and independent research projects.
- Lead teaching assistant, OR 6205: *Deterministic Operations Research*.

## SERVICE

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### Professional

Session Chair, Cooperation and Partnership in Public Services, INFORMS Annual Meeting, 2025

Session Chair, Border Security, Decision Sciences Institute Annual Meeting, 2020

### University and Departmental

Vice President and Treasurer, INFORMS Student Chapter, Northeastern University, 2022 to 2023, 2025 to present  
Instructor and Organizer, Notetaking Workshop for graduate students, INFORMS Student Chapter, Northeastern University, 2024

Secretary, Women in STEM Programs, University of Texas Rio Grande Valley, 2020 to 2021

LEAP Educator, Center for Diversity and Inclusion, University of Texas Rio Grande Valley, 2019

### Community

Board Member, Personnel Committee and Nominating Committee, Beacon Hill Friends House, Boston, MA, 2021 to present

Volunteer Art Teacher, AIESEC Volunteer Program, Mendoza, Argentina, 2014

## PROFESSIONAL EXPERIENCE

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**COMLURB (municipal sanitation company)**, Rio de Janeiro, Brazil, 2014 to 2016

*Financial Analyst*

Secured 100% budget approval at annual town hall meetings by presenting complex financial analysis to non-technical stakeholders. Reduced contract and inventory costs by 30% through auditing and a revised inventory policy. Built analytical frameworks that informed executive strategic planning.

**MRS Global (relocation services)**, Rio de Janeiro, Brazil, 2012 to 2013

*Financial Administrative Assistant*

Designed automated validation protocols for financial control databases, reducing error-correction workload by 60%. Streamlined financial workflows across Excel, TOTVS ERP, and Access.

**Nano Presentes (design manufacturing and retail)**, Rio de Janeiro, Brazil, 2011 to 2012

*Owner and Designer*

Founded and ran a small design business; secured national blog and press coverage.

**Sulamerica Seguros S.A. (insurance)**, Rio de Janeiro, Brazil, 2010 to 2012

*Customer Service Representative*

Handled vehicle accident claims and negotiated emergency service provision.

**Polo Nautico, Universidade Federal do Rio de Janeiro**, Rio de Janeiro, Brazil, and Leeuwarden, Netherlands, 2008

*Team Leader, Naval Design and Engineering Team*

Led the design and construction of a solar-powered boat; secured university and industry sponsorship; earned national and international press coverage.

## TECHNICAL SKILLS

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**Programming and analysis:** Python (6+ years: Pandas, NumPy and SciPy, scikit-learn, NetworkX), R

**Optimization and modeling:** Gurobi, AMPL, LINDO

**Reproducibility and tools:** Git and GitHub, LaTeX, Overleaf, Obsidian

**Teaching:** backward design, active learning, project-based learning, transparent assessment, inclusive pedagogy, curriculum development, online and hybrid course delivery

## LANGUAGES

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Portuguese (native) · English (fluent) · Spanish (conversational)

## PROFESSIONAL MEMBERSHIPS

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Institute for Operations Research and the Management Sciences · Decision Sciences Institute · Tau Beta Pi